

Exploring GAN Variants for Balancing Imbalanced Datasets

1. Introduction

Class imbalance is a common problem in many real-world machine learning applications, especially in domains such as fraud detection, medical diagnosis, and anomaly detection. In imbalanced datasets, one class (the majority class) contains significantly more samples than the other class (the minority class). As a result, machine learning models often become biased toward the majority class and fail to correctly predict minority class instances.

Credit card fraud detection is a typical example of this problem. Fraudulent transactions occur very rarely compared to legitimate ones, but correctly identifying them is extremely important. Traditional classification models trained on such data usually achieve high accuracy while performing poorly on fraud detection, since predicting all samples as legitimate still gives a high accuracy.

In this project, Generative Adversarial Networks (GANs) are used to address the class imbalance problem by generating synthetic samples for the minority class. The performance of a classifier is evaluated before and after applying GAN-based data augmentation. Three GAN models are implemented and compared: Vanilla GAN, Conditional GAN (CGAN), and Wasserstein GAN (WGAN).

2. Problem Statement

The main problem addressed in this project is the poor performance of classification models when trained on highly imbalanced datasets. Since the minority class contains very few samples, the model does not learn enough information to correctly identify it. This leads to low recall and F1-score for the minority class, even when overall accuracy appears high.

The objective of this project is to investigate whether GAN-based data augmentation can help reduce this problem by generating realistic synthetic minority class samples and improving classification performance.

3. Dataset Description

The dataset used in this project is the **Credit Card Fraud Detection dataset**, which contains credit card transactions labeled as either legitimate or fraudulent. The dataset consists of numerical features, where most features have been transformed using Principal Component Analysis (PCA) for privacy reasons.

- **Class 0:** Legitimate transactions
- **Class 1:** Fraudulent transactions

The dataset is extremely imbalanced, with fraud transactions representing approximately **0.17%** of the total data. This severe imbalance makes it an ideal dataset for evaluating data augmentation techniques such as GANs.

4. Imbalance Analysis

To analyze the class imbalance, the distribution of the target classes was visualized using bar charts. The visualization shows that the number of legitimate transactions is vastly greater than the number of fraudulent ones. A logarithmic scale was also used to better display the minority class.

This imbalance explains why traditional classifiers struggle to detect fraud cases and highlights the need for techniques that can increase the number of minority class samples.

5. Methodology Overview

The methodology followed in this project consists of the following steps:

1. Split the original dataset into training and testing sets.
2. Normalize the features using Min-Max scaling.
3. Extract only the minority class samples from the training set.
4. Train GAN models using only the minority class data.
5. Generate synthetic fraud samples using each GAN.
6. Balance the training dataset by adding the generated samples.
7. Train a classifier on the balanced training dataset.
8. Evaluate the classifier on the untouched test dataset.

This approach avoids data leakage and ensures a fair evaluation of the model performance.

6. Generative Adversarial Networks

6.1 Vanilla GAN

The Vanilla GAN consists of two neural networks: a **generator** and a **discriminator**. The generator attempts to generate synthetic samples that resemble real minority class data, while the discriminator tries to distinguish between real and fake samples. Both networks are trained in an adversarial manner, where the generator improves by trying to fool the discriminator.

In this project, the Vanilla GAN was trained using only the minority class samples from the training set. The generator takes random noise as input and outputs synthetic feature vectors, while the discriminator outputs a probability indicating whether a sample is real or fake.

Although Vanilla GANs are simple to implement, they can suffer from unstable training and mode collapse, which may affect the quality of generated samples.

6.2 Conditional GAN (CGAN)

The Conditional GAN extends the Vanilla GAN by incorporating label information into both the generator and the discriminator. This allows the generator to produce samples conditioned on a specific class label.

In this project, the CGAN was conditioned on the fraud class label. The generator received both random noise and the class label as input, while the discriminator evaluated the authenticity of samples together with their labels. This additional information helps guide the generation process and improves sample quality.

CGANs generally provide more controlled and stable generation compared to Vanilla GANs, especially when class information is important.

6.3 Wasserstein GAN (WGAN)

The Wasserstein GAN uses a different loss function based on the Wasserstein distance instead of the traditional binary cross-entropy loss. The discriminator in WGAN, referred to as the **critic**, does not output probabilities but instead estimates how real or fake a sample is.

WGANs are known for more stable training and reduced mode collapse. Weight clipping is applied to the critic to enforce the Lipschitz constraint required by the Wasserstein distance.

In this project, the WGAN was trained only on minority class samples and was used to generate synthetic fraud transactions. Compared to Vanilla GAN, WGAN showed smoother training behavior and more stable convergence.

7. Data Augmentation Process

After training each GAN model, synthetic fraud samples were generated. The number of generated samples was chosen to balance the number of fraud and legitimate transactions in the training set.

Importantly, data augmentation was applied **only to the training set**, while the test set remained unchanged. This ensured that evaluation results reflect real-world performance and are not affected by data leakage.

8. Classification Model

A neural network classifier was used to evaluate the effectiveness of GAN-based data augmentation. The classifier consisted of fully connected layers with dropout regularization to reduce overfitting.

The classifier was trained separately on:

1. The original imbalanced training dataset
 2. The dataset balanced using Vanilla GAN
 3. The dataset balanced using CGAN
 4. The dataset balanced using WGAN
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9. Evaluation Metrics

The performance of the classifier was evaluated using the following metrics:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**

- **Confusion Matrix**

These metrics provide a comprehensive evaluation, especially for imbalanced classification problems where accuracy alone can be misleading.

10. Results and Discussion

The results showed that training on the original imbalanced dataset led to poor recall for the fraud class, despite high accuracy. After applying GAN-based data augmentation, recall and F1-score improved significantly.

Among the three GAN models, the WGAN generally provided the most stable improvements, followed by the CGAN. The Vanilla GAN also improved performance compared to the baseline but showed more variability.

These results indicate that GAN-based data augmentation can effectively enhance minority class detection, especially when combined with proper evaluation practices.

11. Conclusion

In this project, three different GAN models were implemented to address the class imbalance problem in credit card fraud detection. By generating synthetic minority class samples and balancing the training dataset, the performance of the classification model improved significantly.

The comparison between Vanilla GAN, CGAN, and WGAN demonstrated that advanced GAN variants such as WGAN offer more stable training and better performance improvements. Overall, GAN-based data augmentation proved to be an effective approach for handling imbalanced datasets.

12. Future Work

Future work could explore more advanced GAN architectures such as WGAN-GP or apply GAN-based augmentation to other imbalanced datasets. Additionally, experimenting with different classifiers and feature selection techniques may further improve performance.